Experiment 2.2

3D CAD Modelling of a Bracket Component

Aim

To create a 3D bracket component in CAD software by drawing the dimensioned 2D profile and converting the sketch into a solid model using extrusion and related modelling operations.

Requirements

- Computer with CAD software capable of 2D sketching and 3D solid modelling
- Mouse and keyboard
- The supplied dimensioned sketch/reference images

Procedure

1. Create the 2D sketch and enter dimensions. On the appropriate sketch plane, draw the obround base with two mounting holes, the central upright support, the sloping web, and the upper circular boss. Use the dimensions shown in the reference: base length 56 mm between the indicated hole centres, end radius R18 mm, base mounting holes Ø16 mm, upper boss outer diameter Ø30 mm and inner diameter Ø20 mm, and the 40 mm width shown for the upright region. The sketch also shows a 10 mm vertical offset. Constrain and align the features as indicated in the reference drawing.

2. Check the sketch. Verify that the two mounting holes are placed symmetrically in the base and that the upper boss and tapered support align with the central upright before creating solids.

Sketch

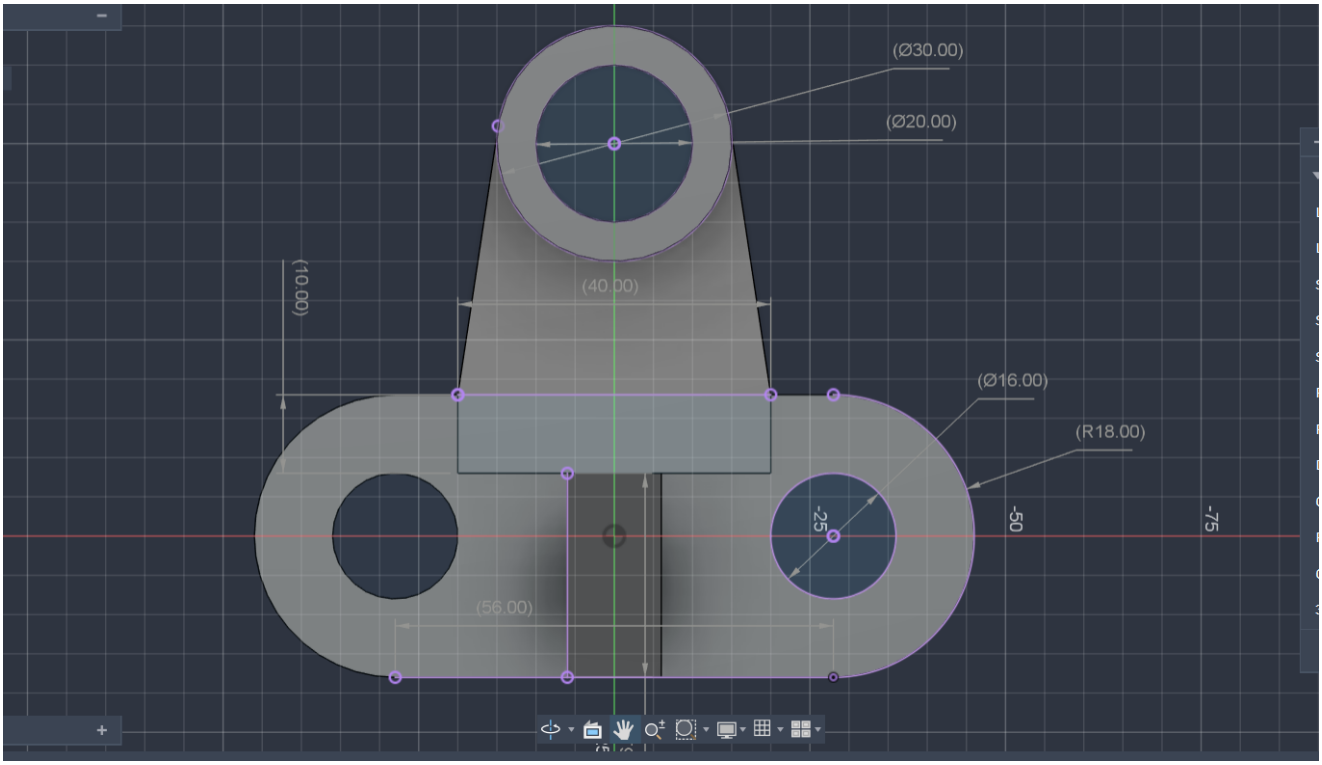


Figure 1. Dimensioned 2D sketch with the principal sizes and hole diameters.

Procedure (continued)

3. Extrude the profiles. Select the closed base and support profiles, then apply the required extrusion thickness to create the solid bracket. Extrude the upper annular profile to form the cylindrical boss, keeping the Ø20 mm opening through the centre. Create the two Ø16 mm mounting holes in the base as shown.

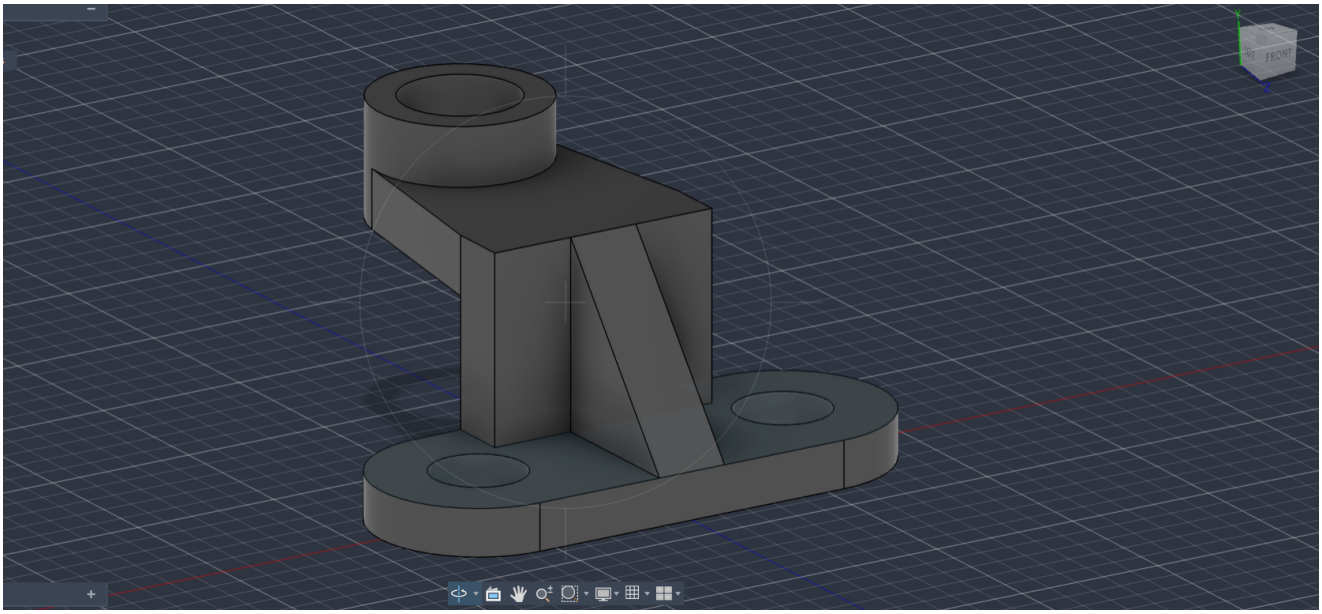


Figure 2. Extruded 3D bracket model showing the base, support web, upright, and upper boss.

4. Inspect the model. Check the front view and isometric view to confirm the shape, hole placement, boss opening, and sloping web. Make any necessary corrections before preparing the drawing sheet.

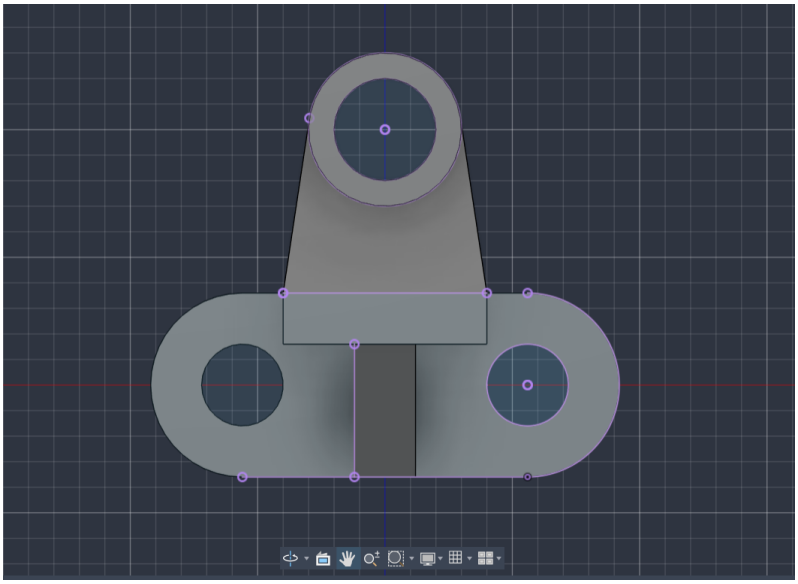


Figure 3. Front view used to inspect the model geometry.

Output

The final output is the completed engineering drawing sheet shown below (the last supplied picture). It presents the bracket in multiple isometric and orthographic views.

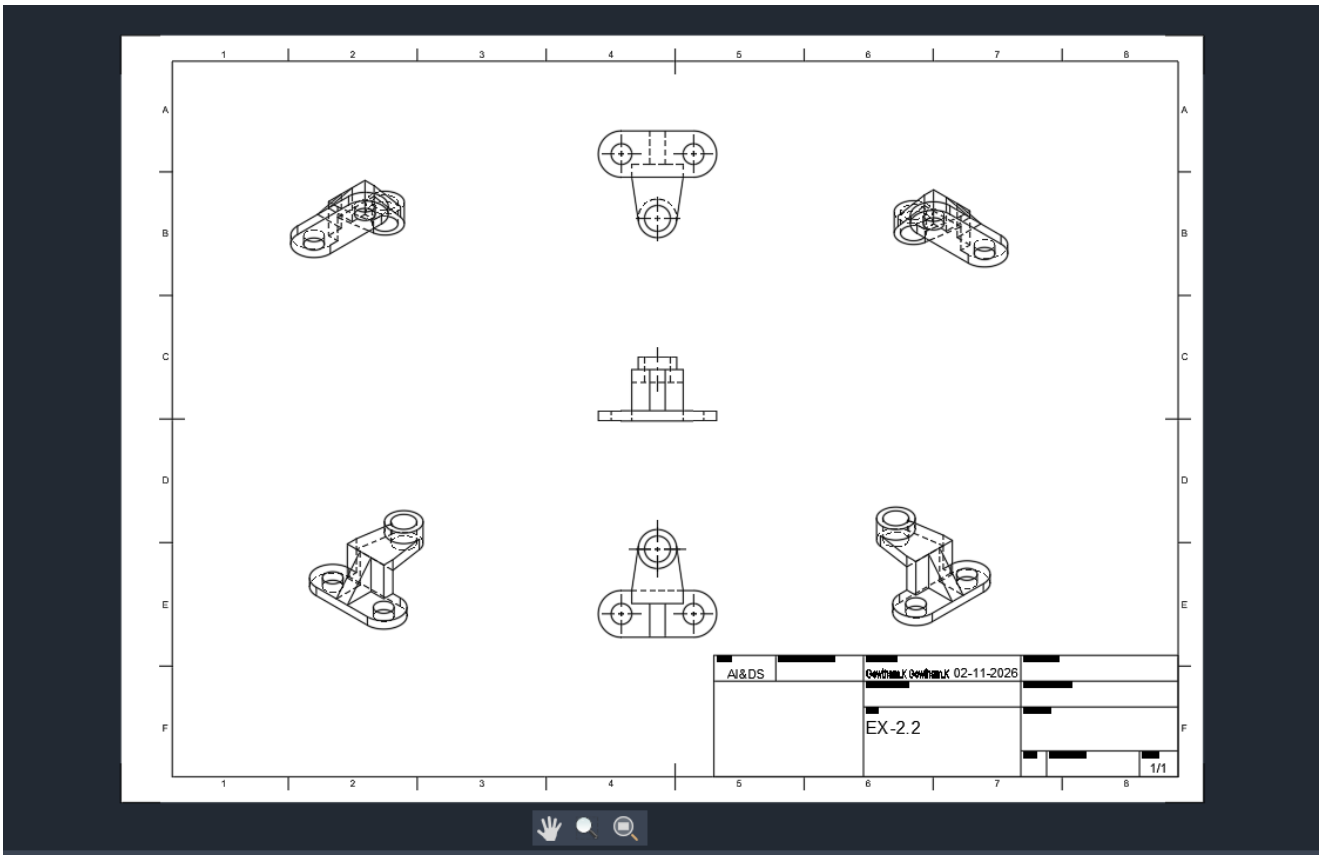


Figure 4. Final drawing sheet showing the completed Experiment 2.2 model and its views.

Result

The 3D bracket component was modelled from the dimensioned sketch using sketching and extrusion operations. The completed drawing sheet shows the final geometry in multiple views, including the mounting holes, upright support, sloping web, and upper boss.